



TSN HOST



Outline

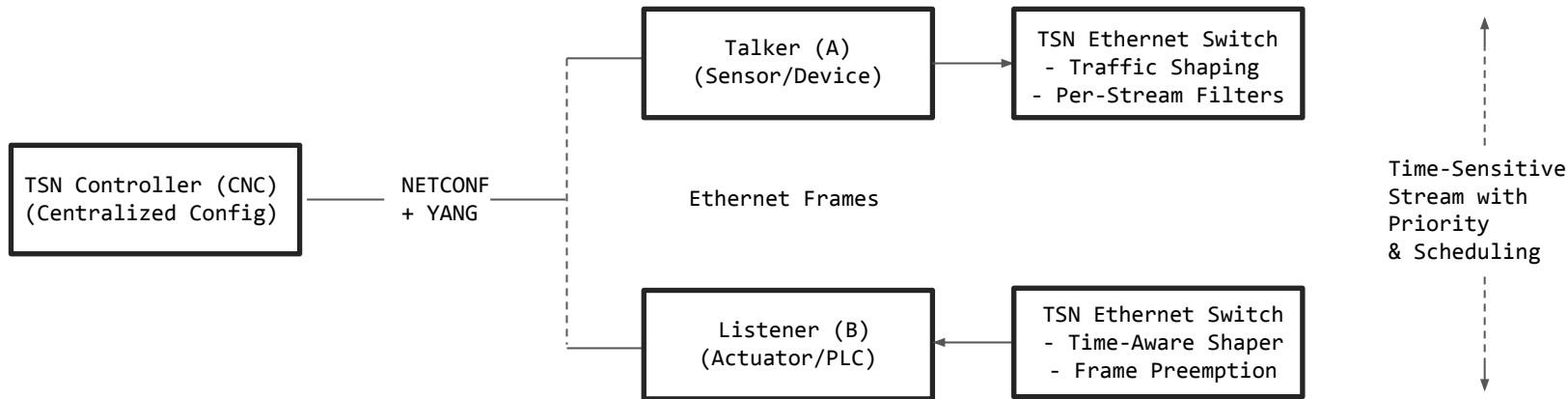
- Project Overview
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 - Hello World
 - Oven Module
- Real-Time UDP Communication
- Sysrepo Plugins and Yang Subscription



Project Overview

- In modern industrial networks, **Time-Sensitive Networking (TSN)** plays a key role in ensuring **deterministic, low-latency communication** for real-time applications such as robotics, automotive systems, and factory automation.
- While the network infrastructure (like switches) is often managed through SDN-based TSN controllers, **end devices** (called **Talkers** and **Listeners**) have traditionally been outside of this centralized control.
- This project focuses on bringing TSN end-stations—**the talkers and listeners**—into the picture by enabling **precisely-timed, UDP-based communication** between them on Linux hosts. We implemented **custom event handlers** to control and monitor real-time data flow.
- By generating periodic UDP streams with accurate transmission timestamps and capturing precise reception times, we effectively simulate the behavior of TSN-capable end devices.

Time-Sensitive Networking(TSN)





Tools and Technologies

- Languages & Environment:
 - **C** – Core implementation of send_udp, rec_udp, oven_event_handler.c and hello_world_event_handler.c
 - **Linux** – Used for low-level socket APIs and timestamping features
 - **Docker** – To containerize and manage the environment
- Configuration & Management:
 - **Netopeer2** – NETCONF server for the TSN host
 - **Sysrepo** – YANG model-based configuration datastore
 - **YANG** – For modeling configuration data structures
- Testing & Deployment:
 - **Command Line Tools** – CLI-based triggering and control
 - **Remote SSH Access** – To start and monitor apps

Event Handling Examples



Basic Event Handling – Hello World

- This simple example verifies that the event-handling mechanism is working. When the configuration is updated through the NETCONF client (edit-config), the Sysrepo subscription triggers a C-based event handler that logs a message.
- Triggered by: `> edit-config --target running --config=/tmp/example-config.xml`
- Output:

```
root@81cf272efc63:/sysrepo# ./hello_world_event_handler
Subscribed to changes in example-module. Waiting for events...
Hello World! Change detected in module: example-module
```
- Event-driven logic from **YANG** → **Sysrepo** → **Handler** is confirmed

Custom Event Handler – Oven Module

- The custom Oven YANG model defines configurable parameters such as temperature and power state. The oven_plugin application listens for configuration changes in Sysrepo. When an update is made via the NETCONF client, the handler reads the new values for temperature and state, and reacts accordingly—such as printing a message or simulating device behavior. This setup effectively simulates how a real device (like a smart oven) might behave.
- Triggered by: `> edit-config --target running --config=/tmp/oven-config.xml`
- Shows how to read and act on real data from YANG
- Confirms that event handling scales beyond Hello World

```
[root@81cf272efc63:/sysrepo# ./oven_plugin
🔍 Starting Oven Plugin...
🔥 [EVENT] Configuration Change Detected 🔥
🔧 New Temperature Set: 0°C
🔧 Oven is OFF
✅ Oven Plugin Ready!
🔥 [EVENT] Configuration Change Detected 🔥
🔧 New Temperature Set: 40°C
🔧 Oven is OFF
```


Real-Time Communication



Real-Time UDP Handler

- Implemented two C-based programs to simulate a TSN host:
 - A **UDP sender** that transmits packets with precise timestamps using **SO_TXTIME**.
 - A **UDP receiver** that listens for packets and logs their **reception timestamps** using **SO_TIMESTAMPING**
- **Sender (send_udp) Highlights:**
 - Sends periodic UDP packets (intervals defined by min/max delay).
 - Each packet includes a scheduled **transmit time (TX_TIME)**.
> /tmp/send_udp -i eth0 -d 0.0.0.0 -u 4242 -S
- **Receiver (rec_udp) Highlights:**
 - Captures **reception timestamp (REC_TIME)** and correlates it with transmitted values.
 - Useful for analyzing delay, jitter, and scheduling accuracy.
> /tmp/rec_udp -i eth0 -p 4242 -S

Real-Time UDP Handler (Contd.)

- **Sender Output:**

```
root@bb9f298c91ea:/sysrepo# /tmp/send_udp -i eth0 -d 0.0.0.0 -p 4242 -S  
min 1 max 99  
timestamping is active with the following SO_TIMESTAMPING value: 4202  
  
txtime of 1st packet is: 1742972265000000000  
1742972266000000000  
D=0.0.0.0, DP=4242, S=eth0, SP=42424, P=0, ID=0, COUNT=0, SEND_TIME=1742972264999134903  
D=0.0.0.0, DP=4242, S=eth0, SP=42424, P=0, ID=0, COUNT=1, SEND_TIME=1742972265999323593
```

- **Receiver Output:**

```
root@bb9f298c91ea:/sysrepo# /tmp/rec_udp -i eth0 -p 4242 -S  
Timestamping is active with the following SO_TIMESTAMPING value: 26  
REC_TIME=1742972264999136535, UDP_LENGTH=78, COUNT=0, TX_TIME=1742972265000000000, D=0.0.0.0, DP=4242, S=eth0, SP=42424, P=0, ID=0  
REC_TIME=1742972265999326698, UDP_LENGTH=78, COUNT=1, TX_TIME=1742972266000000000, D=0.0.0.0, DP=4242, S=eth0, SP=42424, P=0, ID=0
```

System Integration & Configuration Trials



Subscriptions to the YANG models

Approaches:

1. Custom plugins
2. Sysrepo-plugins



SYSREPO-PLUGINS

- Configuring linux systems using Sysrepo NETCONF server implementation.
- Collection of plugins based on different YANG modules.
- Plugins :
 - ietf-system-plugin (focused on)
 - ietf-interfaces-plugin
 - ietf-routing-plugin



SYSREPO-PLUGINS

→ Installation:

- Dependencies like libyang, libyang-cpp, sysrepo, sysrepo-cpp etc. are important !

→ Setup:

- Standard C-make build process.
- Installing YANG models.



SYSREPO-PLUGINS

Working :

- Figuring out whether the plugins can be used on custom YANG module.
- Whether the existing code could be modified to do so.

Tasks attempted:

- Created a simple example YANG module and modified the code to monitor the module.
 - **Result:** The ietf-system-plugin uses a registry mechanism. The YANG module has to be entered in that system.
- Worked on ietf-system yang module. Tracking change of hostname.
 - **Result:** The modified the code to print a message. Limitation due to containerized environment.


```

[ERR] ietf-system-plugin: Failed to apply datastore values for the following paths:
[ERR] ietf-system-plugin: /ietf-system:system/clock/timzone-name
[ERR] ietf-system-plugin: Reason: [org.freedesktop.Dbus.Error.FileNotFound] Failed to open bus (No such file or directory)
[INF] ietf-system-plugin: Applying running datastore values for module Dns
[INF] No "running" datastore changes to apply.
[ERR] ietf-system-plugin: Failed to apply datastore values for the following paths:
[ERR] ietf-system-plugin: /ietf-system:system/dns-resolver
[ERR] ietf-system-plugin: Reason: [org.freedesktop.Dbus.Error.FileNotFound] Failed to open bus (No such file or directory)
[ERR] ietf-system-plugin: Applying running datastore values for module Ntp
[INF] No "running" datastore changes to apply.
[ERR] ietf-system-plugin: Failed to apply datastore values for the following paths:
[ERR] ietf-system-plugin: /ietf-system:system/ntp
[ERR] ietf-system-plugin: Reason: [org.freedesktop.Dbus.Error.FileNotFound] Failed to open bus (No such file or directory)
[INF] ietf-system-plugin: Applying running datastore values for module Authentication
[INF] No "running" datastore changes to apply.
[ERR] Unspecified pattern - "$y$9t$43$C$u$W$1d$G$u$1W$1P$Q$S1.$9t$4Q$3$y$u$K$8$9$S$10P$G$M$M" does not conform to "$0$.[${1}[a-zA-Z0-9_/]{1,8}]${2}[a-zA-Z0-9_/]{22}]$0$ (rounds=${4})${3}[a-zA-Z0-9_/]{1,16}]${4}[0-9]{1,16}]${5}[a-zA-Z0-9_/]{80}"; (path "/ietf-system:system/authentication/user/password")
[ERR] Invalid datastore edit.
[ERR] ietf-system-plugin: Failed to apply datastore values for the following paths:
[ERR] ietf-system-plugin: /ietf-system:system/authentication/user
[ERR] ietf-system-plugin: Reason: Session::setUser: Couldn't set /ietf-system:system/authentication/user[name='root']/password to "$y$9t$43$C$u$W$1d$G$u$1W$1P$Q$S1.$9t$4Q$3$y$u$K$8$9$S$10P$G$M$M"; SR_ERR_INVALID_ARG
[ERR] Unspecified pattern - "$y$9t$43$C$u$W$1d$G$u$1W$1P$Q$S1.$9t$4Q$3$y$u$K$8$9$S$10P$G$M$M" does not conform to "$0$.[${1}[a-zA-Z0-9_/]{1,8}]${2}[a-zA-Z0-9_/]{22}]$0$ (rounds=${4})${3}[a-zA-Z0-9_/]{1,16}]${4}[0-9]{1,16}]${5}[a-zA-Z0-9_/]{80}"; (path "/ietf-system:system/authentication/user/password") (SR_ERR_INVALID_ARG)
[ERR] Invalid datastore edit. (SR_ERR_INVALID_ARG)
[INF] ietf-system-plugin: Registering operational callbacks for module System
[INF] ietf-system-plugin: Creating operational subscription for xpath /ietf-system:system-state/platform
[INF] Recovering module "ietf-system" operational get subscription of CID 7.
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[INF] Recovering module "ietf-system" operational get subscription of CID 7.
[INF] ietf-system-plugin: Creating operational subscription for xpath /ietf-system:system-state/clock
[INF] ietf-system-plugin: Registering module change callbacks for module System
[INF] ietf-system-plugin: Registering RC callbacks for module System
[INF] Recovering RC/action "ietf-system:system-restart" subscription of CID 7.
[INF] ietf-system-plugin: Creating RC subscription for xpath /ietf-system:system-shutdown
[INF] Recovering RC/action "ietf-system:system-shutdown" subscription of CID 7.
[INF] ietf-system-plugin: Creating RC subscription for xpath /ietf-system:set-current-datetime
[INF] Recovering RC/action "ietf-system:set-current-datetime" subscription of CID 7.
[INF] ietf-system-plugin: Registered module System
[INF] ietf-system-plugin: Registering operational callbacks for module Hostname
[INF] ietf-system-plugin: Creating operational subscription for xpath /ietf-system:system/hostname
[INF] ietf-system-plugin: Registering module change callbacks for module Hostname
[INF] ietf-system-plugin: Creating module change subscription for xpath /ietf-system:system/hostname
[INF] Recovering module "ietf-system" running change subscription of CID 7.
[INF] Recovering module "ietf-system" running change subscription of CID 7.
[INF] Recovering module "ietf-system" running change subscription of CID 7.
[INF] Recovering module "ietf-system" running change subscription of CID 7.
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[INF] Recovering module "ietf-system" running change subscription of CID 7.
[INF] Recovering module "ietf-system" running change subscription of CID 7.
[INF] ietf-system-plugin: Registering RC callbacks for module Hostname
[INF] ietf-system-plugin: Registered module Hostname
[INF] ietf-system-plugin: Creating operational subscription for xpath /ietf-system:system/clock/timzone-name
[INF] ietf-system-plugin: Registering module change callbacks for module Timzone
[INF] ietf-system-plugin: Creating module change subscription for xpath /ietf-system:system/clock/timzone-name
[INF] ietf-system-plugin: Registering RC callbacks for module Timzone
[INF] ietf-system-plugin: Registered module Timzone
[INF] ietf-system-plugin: Registering operational callbacks for module Dns
[INF] ietf-system-plugin: Creating operational subscription for xpath /ietf-system:system/dns-resolver/search
[INF] ietf-system-plugin: Creating operational subscription for xpath /ietf-system:system/dns-resolver/serve
[INF] ietf-system-plugin: Creating operational subscription for xpath /ietf-system:system/dns-resolver/options
[INF] ietf-system-plugin: Registering module change callbacks for module Dns
[INF] ietf-system-plugin: Creating module change subscription for xpath /ietf-system:system/dns-resolver/search
[INF] ietf-system-plugin: Creating module change subscription for xpath /ietf-system:system/dns-resolver/serve
[INF] ietf-system-plugin: Creating module change subscription for xpath /ietf-system:system/dns-resolver/options
[INF] ietf-system-plugin: Registering RC callbacks for module Dns
[INF] ietf-system-plugin: Registered module Dns
[INF] ietf-system-plugin: Registering operational callbacks for module Ntp
[INF] ietf-system-plugin: Creating operational subscription for xpath /ietf-system:system/ntp
[INF] ietf-system-plugin: Registering module change callbacks for module Ntp
[INF] ietf-system-plugin: Creating module change subscription for xpath /ietf-system:system/ntp/serve
[INF] ietf-system-plugin: Creating module change subscription for xpath /ietf-system:system/ntp/enabled
[INF] ietf-system-plugin: Registering RC callbacks for module Ntp
[INF] ietf-system-plugin: Registered module Ntp
[INF] ietf-system-plugin: Registering operational callbacks for module Authentication
[INF] ietf-system-plugin: Creating operational subscription for xpath /ietf-system:system/authentication/user
[INF] ietf-system-plugin: Registering module change callbacks for module Authentication
[INF] ietf-system-plugin: Creating module change subscription for xpath /ietf-system:system/authentication/user
[INF] ietf-system-plugin: Creating module change subscription for xpath /ietf-system:system/authentication/user/authorized-key
[INF] ietf-system-plugin: Registering RC callbacks for module Authentication
[INF] ietf-system-plugin: Registered module Authentication
[INF] ietf-system-plugin: Created plugin subscriptions
[ERR] Sysrepo-cpp: User callback threw an exception: [org.freedesktop.Dbus.Error.FileNotFound] Failed to open bus (No such file or directory)
[ERR] sysrepo called after throwing an instance of "std::runtime_error"
[ERR] what(): [org.freedesktop.Dbus.Error.FileNotFound] Failed to open bus (No such file or directory)
Aborted (core dumped)
root@63c4d349df9b:/sysrepo-plugins/build/plugins/ietf-system-plugins #

```

```

SYSREPO-PLUGINS [CO...  i  ↻  ⌕  ⌕
plugins > ietf-system-plugin > src > plugin > modules > C++ hostname.cpp
plugins 109 namespace ietf::sys::sub::change {
122 /**
123  *
124  */
138 sr::ErrorCode
139 HostnameModuleChangeCb::operator()(sr::Session session, uint32_t subscriptionId,
140 std::string_view moduleName,
141 std::optional<std::string_view> subXPath,
142 sr::Event event, uint32_t requestId)
143 {
144     sr::ErrorCode error = sr::ErrorCode::Ok;
145
146     Hostname hostname_handle;
147
148     switch (event) {
149     case Sysrepo::Event::Change:
150         for (auto& change : session.getChanges(subXPath->data())) {
151             switch (change.operation) {
152             case Sysrepo::ChangeOperation::Created:
153             case Sysrepo::ChangeOperation::Modified: {
154                 // modified hostname - get current value and use sethostname()
155                 auto value = change.node.asItem().value();
156                 auto hostname = std::get<std::string>(value);
157
158                 try {
159                     hostname_handle.setValue(hostname);
160                     std::cout << "Hostname has been modified to: " << hostname << std::endl;
161                 } catch (const std::runtime_error& err) {
162                     SRPLG_LOG_ERR(ietf::sys::PLUGIN_NAME, "%s", err.what());
163                     error = sr::ErrorCode::OperationFailed;
164                 }
165                 break;
166             }
167             default:
168                 break;
169             }
170         }
171     }
172 }

```

```

root@63c4d349df9b:/# netopeer2-cli
load_config: No saved configuration.
> connect
Keyboard-Interactive Authentication
Please enter your authentication token
root's password:
> edit-config --target running --config=sysrepo-plugins/plugins/ietf-system-plugin/examples/hostname.xml
ERROR

type:      application
tag:       operation-failed
severity:  error
message:   EV ORIGIN: SHM event "rpc" ID 6 processing timed out.

type:      application
tag:       operation-failed
severity:  error
message:   User callback failed.
>

```



Outcome

- The sysrepo-plugins could be used for specific configurations.
- Scope outside containers needs to be looked at.

Talker and Listener applications



Objective:

- Compile and execute the C files.
- Simulation of real-time communication between TSN hosts.

Tasks achieved:

- Compiled the files `rec_udp.c` and `send_udp.c`
- Created virtual interfaces.
- Started the applications `rec_udp` and `send_udp`.

(Note: Applied specific privileges to the container)

```
root@docker-desktop:/# ip link add tsn0 type veth peer name tsn1
root@docker-desktop:/# ip addr add 192.168.1.1/24 dev tsn0
ip addr add 192.168.1.2/24 dev tsn1
root@docker-desktop:/# ip link set tsn0 up
ip link set tsn1 up
root@docker-desktop:/# ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: services1@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP mode DEFAULT group default
    link/ether a2:56:9d:27:f9:aa brd ff:ff:ff:ff:ff:ff link-netnsid 0
3: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN mode DEFAULT group default qlen 1000
    link/ether 26:c2:15:43:79:95 brd ff:ff:ff:ff:ff:ff
4: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN mode DEFAULT group default
    link/ether 02:42:0e:e7:94:7c brd ff:ff:ff:ff:ff:ff
7: tsn1@tsn0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP mode DEFAULT group default qlen 1000
    link/ether 02:11:20:71:73:4f brd ff:ff:ff:ff:ff:ff
8: tsn0@tsn1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP mode DEFAULT group default qlen 1000
    link/ether 42:32:5e:7c:85:ab brd ff:ff:ff:ff:ff:ff
root@docker-desktop:/# cd /usr/local/bin/
root@docker-desktop:/usr/local/bin# rec_udp -i tsn1 -p 4242
Timestamping is active with the following SO_TIMESTAMPING value: 69
```

```
root@docker-desktop:/usr/local/bin# send_udp -i tsn0 -d 192.168.1.2 -u 4242 -S
min 1 max 99
pthread_setschedparam: Operation not permitted
root@docker-desktop:/usr/local/bin# send_udp -i tsn0 -d 192.168.1.2 -u 4242 -n
min 1 max 99
pthread_setschedparam: Operation not permitted
```



Outcome

- The applications are properly compiled and executed.
- The actual code need to be applied in a setup of VLAN interface or physical NIC.



THANK YOU...